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STRETCHABLE SENSOR ASSEMBLY

Abstract

Systems are provided for a device comprising an emitter, a detector, a power/signal cable directly connected to one of the detector or the emitter, and a wave-shaped flexible printed circuit board (PCB) disposed between the emitter and the detector, wherein the wave-shaped flexible PCB includes a first plurality of conductive traces. The device may further comprise an elastic cover which makes the device a stretchable sensor assembly.

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Background/Summary

TECHNICAL FIELD

[0001] Embodiments of the subject matter disclosed herein relate to a stretchable assembly for adjusting a distance between two components of the stretchable sensor assembly.

BACKGROUND

[0002] In some examples a finger pulse oximeter sensor may be used for measuring saturation of oxygen in blood of a patient (SpO2). The finger sensor may include a light emitting diode (LED) emitter adapted to be positioned on a first side of the finger and a detector adapted to be positioned on a second side of the finger, opposite the first side. For accurate oxygen saturation measurements, the emitter and detector are aligned such that light from the emitter passes through the finger and interacts with the detector. When the finger is a larger or smaller diameter than expected, the emitter and detector may become misaligned, leading to inaccurate determination of oxygen saturation in the blood.

BRIEF DESCRIPTION

[0003] In one embodiment, a device, comprising: an emitter; a detector; a power/signal cable directly connected to one of the detector or the emitter; and a wave-shaped flexible printed circuit board (PCB) disposed between the emitter and the detector, wherein the wave-shaped flexible PCB includes a first plurality of conductive traces.

[0004] It should be understood that the brief description above is provided to introduce in simplified form a selection of concepts that are further described in the detailed description. It is not meant to identify key or essential features of the claimed subject matter, the scope of which is defined uniquely by the claims that follow the detailed description. Furthermore, the claimed subject matter is not limited to implementations that solve any disadvantages noted above or in any part of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] The present invention will be better understood from reading the following description of non-limiting embodiments, with reference to the attached drawings, herein below:

[0006] FIG. **1** shows an example from the prior art of a non-stretchable sensor assembly on a large subject.

[0007] FIG. **2** shows an example from the prior art of the non-stretchable sensor assembly on a small subject.

[0008] FIG. **3** shows an example of a stretchable sensor assembly fitted to a large subject.

[0009] FIG. **4** shows an example of the stretchable sensor assembly fitted to a small subject.

[0010] FIG. **5** shows a perspective view of hardware of a stretchable sensor assembly in a compact state.

[0011] FIG. **6** shows the perspective views of hardware of the stretchable sensor assembly in a stretched state.

[0012] FIG. **7** shows a perspective view of a stretchable sensor assembly in a compact state.

[0013] FIG. **8** shows a side view of the stretchable sensor assembly in the compact state.

[0014] FIG. **9** shows a side view of the stretchable sensor assembly in a stretched state.

[0015] FIG. **10** shows a first view of the stretchable sensor assembly positioned on a finger.

[0016] FIG. **11** shows a second view of the stretchable sensor assembly positioned on the finger.

[0017] FIG. **12** shows a block diagram illustrating an example pulse oximetry system.

DETAILED DESCRIPTION

[0018] The following description relates to a stretchable sensor assembly. In an exemplary embodiment, the stretchable sensor is a finger pulse oximetry sensor, however other types of sensors are also considered. A sensor, such as a finger pulse oximetry sensor, may include two components which demand a relative alignment in order to sense with reduced noise and power consumption, and thus produce an accurate reading. In examples of the prior art, shown in FIGS. **1-2**, the components of a non-stretchable sensor assembly may become unaligned when fitted to a target, such as finger, which is either larger (see FIG. **1**) or smaller (see FIG. **2**) than the intended target size. The issues shown in FIGS. **1-2** may be at least partially addressed by a stretchable sensor assembly. The stretchable sensor assembly may be fitted in a stretched state when the target is large, as shown in FIG. **3** or may be fitted in a compact state when the target is small, as shown in FIG. **4**.

[0019] In the exemplary embodiment where the stretchable sensor is a finger pulse oximetry sensor such as shown as a schematic representation in FIG. **12**, the sensor may include hardware as shown in FIG. **5-6**. The hardware may include components affixed to rigid-flexible printed circuit board (PCB) and/or flexible PCB with stiffener positioned in some portions. A portion of the sensor assembly electrically coupling the two components may be a flexible PCB and may be adapted to be stretchable. In this way, the two components are both electrically coupled via each other to a power/signal cable, and separate couplers are not demanded for each of the two components such that a single coupler may electrically couple the two components or a plurality of components the same configuration as the two components. By not demanding separate couplers, the sensor assembly is compact and is not prone to snagging on other hospital equipment. The hardware may be further covered by an elastic sleeve to form the stretchable sensor assembly shown in FIGS. **7-9**. The stretchable sensor assembly may be positioned around a finger to sense saturation of oxygen in the blood as shown in FIGS. **10-11**.

[0020] Turning now to FIG. **1**, it shows a non-stretchable sensor assembly **100** of the prior art. The non-stretchable sensor assembly includes a first component **102** affixed to a first end **104** of a body **106**. A second component **108** may be affixed towards a second end **110** of the non-stretchable sensor assembly **100**. The body **106** may be non-stretchable, holding the first component **102** at a fixed distance **120** (along a length of the body **106**) from the second component **108**. The non-stretchable sensor assembly may be positioned on a large target **112**. As one example, the large target **112** may be circular and may have a diameter **114**.

[0021] In an example where the non-stretchable sensor assembly **100** is a finger pulse oximeter, the first component **102** may be an LED array adapted to emit collimated light **118** and the large target **112** may be a finger. The collimated light **118** may travel through the finger. When positioned on a medium sized finger with a diameter smaller than diameter **114**, collimated light **118** may pass through the finger and hit the second component **108** which may be a photodetector. In this way, light may pass through the finger, interact with hemoglobin in the blood and be detected by the detector to determine SpO2. However, when diameter **114** is large as is shown in FIG. **1**, the body **106** may experience a shape change to fit over the large diameter, including deflection in a radial direction from the large target **112**, causing the detector (e.g., second component **108**) to not be positioned directly in the path of collimated light **118**. Consequently, an accuracy of the determined SpO2 may be decreased.

[0022] The first component and second component may also be unaligned if the small target **204** has a small diameter **202** as shown in FIG. **2**. The small diameter **202** may be smaller than a medium sized target (e.g., finger). Shape change to the body **106** to fit small diameter **202** may also cause collimated light **118** to be misaligned with the detector (e.g., second component **108**). For example, when misaligned, collimated light **118** or another beam projected from the first component **102** may not be centered along an axis to touch an approximate detection point of the second component **108** or be outside a cone of detection extending radially from the detection point of the second component **108**. A non-stretchable sensor array may be provided in more than one size, to account for different sized targets. However, keeping a stock of different sized sensors increases a cost and complexity of using the sensor in a setting, such as a hospital setting, where patients of many different sizes are seen.

[0023] A stretchable sensor assembly, such as stretchable sensor assembly **300** shown in FIGS. **3** and **4** may be used as a one size fits all sensor assembly. The stretchable sensor assembly **300** may be used more accurately than the non-stretchable sensor assembly of FIGS. **1** and **2** with both a small target and a large target. FIG. **3** shows the large target **112** of FIG. **1** fitted with a stretchable sensor assembly **300** in a stretched configuration. Stretchable sensor assembly **300** may include first component **102** and second component **108**. Each of first component **102** and second component **108** of stretchable sensor assembly **300** may be affixed to opposite ends of a stretchable band **302**. As shown in FIG. **3**, stretchable band **302** may be in a fully stretched state, thereby extending a distance between first component **102** and second component **108**. In this way, first component **102** and second component **108** may be aligned. In the example where stretchable sensor assembly **300** is a finger pulse oximetry sensor, collimated light **118** from the emitter may reach the detector and SpO2 may be determined more accurately than when collimated light **118** is misaligned from the detector.

[0024] Stretchable band **302** may be shown in a compact state in FIG. **4**, in which stretchable band **302** is collapsed compared to the stretched state shown in FIG. **3**. The compact state may include stretchable band **302** being collapsed or folded into an accordion shape or a wave shape, thereby decreasing the distance between first component **102** and second component **108**. Thus, first component **102** may be aligned with second component **108** when fitted to small target **204**. For example, when the stretchable band **302** stretched or collapsed/folded, the first component **102** and second component **108** may remain approximately centered on the same axis as before the stretchable band **302** is stretched or folded. The collimated light **118** may be projected approximately along or in parallel with the axis.

[0025] In contrast with the non-stretchable sensor assembly **100** of prior art shown in FIGS. **1** and **2**, the stretchable sensor assembly **300** may fit both large target **112** and small target **204** such that first component **102** and second component **108** may be aligned. In this way, the stretchable sensor assembly **300** may fit a range of target sizes (e.g., a range between and including sizes of small target **204** and large target **112**) and prevent misalignment of first component **102** and second component **108**. In some examples, stretchable band **302** may be elastic, such that the compact state may be a resting state of stretchable band **302**. The stretchable band **302** may return to the compact state when not under tension, and the stretched state may be reached by applying tension to stretchable band **302**. The stretchable band **302** may have a plurality of stretched states, where the increasing tension between a first stretched state and a second stretched state may increase the distance the stretchable band **302** is stretched. Thus, stretchable band **302** may experience shape change from the compact state, or resting state, to accommodate larger target sizes and promote alignment of first component **102** and second component **108**. In the example where stretchable sensor assembly **300** is a finger pulse oximeter, first component **102** may be a light emitter that emits light which may be received by second component **108** due to the alignment thereof after traveling through the target, for a range of target sizes (e.g., a range of finger sizes).

[0026] As described above, in some embodiments, a stretchable sensor assembly may be a finger pulse oximetry sensor. FIG. **12** shows a block diagram of one embodiment of a pulse oximetry system **1220**. Light transmitted from an emitter unit **1200** passes into patient tissue **1202**. The emitter unit includes multiple light sources **1201**, such as light-emitting diodes (LEDs), with each light source having a dedicated wavelength. Each wavelength forms one measurement channel on which photoplethysmography (PPG) waveform data are acquired.

[0027] The light transmitted through the patient tissue **1202** is received by a detector unit **1203**, which comprises two photodetectors **1204** and **1205** in this example. For example, photodetector **1204** may be a silicon photodiode, and photodetector **1205** may be a second silicon photodiode with different spectral characteristics or an indium gallium arsenide (InGaAs) photodiode. The emitter and detector units form a probe subunit **1223** of the pulse oximetry system **1220**.

[0028] The probe subunit **1223** may be coupled (e.g., electrically and/or communicatively) to a drive and processing subunit **1221** via a cable **1207** and one or more connectors. For example, a connector may be present on an end of cable **1207** to connect cable **1207** and probe subunit **1223** to drive and processing subunit **1221**. In this way, probe subunit **1223** may be removably coupled to drive and processing subunit **1221**, such that probe subunit **1223** may be communicatively coupled

and decoupled from the drive and processing subunit **1221**. In other examples, probe subunit **1223** and drive and processing subunit **1221** may be integrated into the same housing.
[0029] Drive and processing subunit **1221** may include an input amplifier unit **1206** and an emitter drive unit **1208**. The photodetectors convert the optical signals received into electrical pulse trains and feed them to an input amplifier unit **1206**. The amplified measurement channel signals are further supplied to a control unit **1210**, which executes instructions stored in memory **1212** to convert the signals into digitized format for each wavelength channel.
[0030] The control unit **1210** further controls emitter drive unit **1208** to alternately activate the light sources. To activate the light sources, the emitter drive unit **1208** may include a voltage source, such as a battery, described in more detail below. As mentioned above, each light source is typically illuminated several hundred times per second. With each light source being illuminated at such a high rate compared to the pulse rate of the patient, the control unit **1210** obtains a high number of samples at each wavelength for each cardiac cycle of the patient. The value of these samples varies according to the cardiac cycle of the patient, where the variation may be caused by the arterial blood.
[0031] The digitized PPG signal data at each wavelength may be stored in memory **1212** of the control unit **1210** before being processed further according to non-transitory instructions (e.g., algorithms) executable by the control unit **1210** to obtain physiological parameters. Memory **1212** may comprise a suitable data storage medium, for example, a permanent storage medium, removable storage medium, and the like. Additionally, memory **1212** may be a non-transitory storage medium. In some examples, the pulse oximetry system **1220** may include a communication subsystem **1217** operatively coupled to one or more remote computing devices, such as hospital workstations, smartphones, and the like. The communication subsystem **1217** may enable the output from the detector units (e.g., the digitized PPG signal data) to be sent to the one or more remote computing devices for further processing, and/or the communication subsystem may enable the output from the algorithms discussed below (e.g., determined physiological parameters) to be sent to the remote computing devices. The communication subsystem **1217** may include wired and/or wireless communication devices compatible with one or more different communication protocols. As non-limiting examples, the communication subsystem **1217** may be configured for communication via a wireless telephone network, a local-area or wide-area network, and/or the Internet.
[0032] Algorithms may utilize the same digitized signal data and/or results derived from the algorithms and stored in the memory **1212**, for example. For example, to determine oxygen saturation and pulse transit time (PTT), the control unit **1210** may be adapted to execute an SpO2 algorithm and a PTT algorithm, respectively. For this example, the SpO2 algorithm and a PTT algorithm may be stored in the memory **1212** of the control unit **1210**. Additional algorithms, such as a blood pressure algorithm, a hypovolemia algorithm, and a respiration rate algorithm, may also be stored in memory **1212** for determining blood pressure, an indication of hypovolemia, and respiration rate, respectively. The obtained physiological parameters and waveforms may be shown on a screen of a display unit **1214**. Further, in some examples, the control unit, memory, and/or other subsystems may be located remotely from the rest of the sensor on a separate device, and the signal data from the detector units may be sent to the separate device for processing.
[0033] The input amplifier unit **1206**, the control unit **1210** and memory **1212**, the emitter drive unit **1208**, probe subunit **1223**, and/or additional components (the display unit, for example) may collectively form a sensor. As used herein, the term “probe” may refer to the probe and the attachment parts that attach the optical components of the probe to the tissue site. The term pulse oximeter or pulse oximetry sensor may refer to a unit comprising a probe, an analog front end, and a signal processing unit that calculates SpO2 and other blood characteristics. In a multi-parameter body area network system, the system typically represents a set of multiple sensors, e.g., the different physiological parameter measurements. Therefore, the whole measurement system may

comprise several sensors, and the sensors may communicate to a common hub in which the parameters' information is integrated.

[0034] As used herein, the terms “sensor,” “system,” “unit,” or “module” may include a hardware and/or software system that operates to perform one or more functions. For example, a sensor, module, unit, or system may include a computer processor, controller, or other logic-based device that performs operations based on instructions stored on a tangible and non-transitory computer readable storage medium, such as a computer memory. Alternatively, a sensor, module, unit, or system may include a hard-wired device that performs operations based on hard-wired logic of the device. Various modules or units shown in the attached figures may represent the hardware that operates based on software or hardwired instructions, the software that directs hardware to perform the operations, or a combination thereof.

[0035] “Systems,” “units,” “sensors,” or “modules” may include or represent hardware and associated instructions (e.g., software stored on a tangible and non-transitory computer readable storage medium, such as a computer hard drive, ROM, RAM, or the like) that perform one or more operations described herein. The hardware may include electronic circuits that include and/or are connected to one or more logic-based devices, such as microprocessors, processors, controllers, or the like. These devices may be off-the-shelf devices that are appropriately programmed or instructed to perform operations described herein from the instructions described above. Additionally or alternatively, one or more of these devices may be hard-wired with logic circuits to perform these operations.

[0036] An example of pulse oximetry hardware **500** is shown in FIGS. **5-6**. Pulse oximetry hardware **500** may include the probe subunit **1223** and the connect cable **1207** of FIG. **12**. Additionally or alternatively, the pulse oximetry hardware may be incorporated into a stretchable pulse oximetry sensor assembly, such as stretchable sensor assembly **700** as shown in FIGS. **7-11**. Reference axes **501** are provided for comparison between the views shown in FIGS. **5-9**, including an x-axis, a y-axis, and a z-axis.

[0037] Pulse oximetry hardware **500** may include a cable **502** (e.g., the connect cable **1207** of FIG. **12**). Cable **502** may couple pulse oximetry hardware **500** to a power source (not shown), supplying power to components of pulse oximetry hardware **500**. Additionally or alternatively, cable **502** may communicatively couple pulse oximetry hardware **500** to a controller (e.g., drive and processing subunit **1221** of FIG. **12**) such that the pulse oximetry hardware may send and receive signals from the controller. The controller may include instructions to emit light and interpret signals from a detector to provide a user with a quantitative SpO2 of a patient wearing pulse oximetry hardware **500**, for example around the finger of the patient or another example of the large target **112** or the small target **204** of FIGS. **1-4**. Cable **502** may be a power/signal cable which includes a plurality of wires **504**. In other examples, cable **502** may be a battery and transmission unit that allows for wireless operation of pulse oximetry hardware **500**. The plurality of wires **504** may be mounted to a first rigid/stiffener portion **503**, where first rigid/stiffener portion **503** may be a rigid portion of a rigid-flex-rigid PCB or a stiffener of a flex PCB with stiffeners. In this way, the rigidity of first rigid/stiffener portion **503** may prevent damage to the connections (e.g., solder joints) between the plurality of wires **504** and first rigid/stiffener portion **503**. In some examples, first rigid/stiffener portion **503** may be rectangular with side extensions on which a support **507** of cable **502** is connected for physical support to the couplings between the plurality of wires **504** and first rigid/stiffener portion **503**. The support **507** may wrap around cable **502**, and be in face sharing contact with first rigid/stiffener portion **503**. For an example, the support **507** may curve around the cable **502**, such as to be positioned radially about the cable **502**. The support **507** may be a sleeve or include a sleeve component. The support **507** may physically couple and/or connect to the first rigid/stiffener portion **503** via a plurality of appendages **509**. The appendages **509** may be part of the support **507**, such as being stamped from or molded with the support **507**. The appendages **509** may be connected and/or physically coupled to the support **507**, such as via joining. For an example there may be two of the appendages **509**, with a first appendage opposite to and mirroring a second appendage across the support **507**. In examples wherein pulse oximetry hardware **500** includes a transmission unit and a battery unit rather than wires in a power/signal cable, the transmission unit and battery unit may be coupled to the first rigid/stiffener portion **503**.

[0038] First rigid/stiffener portion **503** may physically couple to a first end of a first flex portion **506**, where first flex portion **506** is located opposite the plurality of wires across first rigid/stiffener portion **503**, and a second end of first flex portion **506** may physically couple to a second rigid/stiffener portion **510**. First flex portion **506** may be a flex portion of a flex PCB with stiffeners, or a flex portion of a rigid-flex-rigid PCB. First flex portion **506** may be ribbon shaped with a thin thickness (e.g., dimension in the z-direction), and a longer length (e.g., dimension in the y-direction, between first rigid/stiffener portion **503** and second rigid/stiffener portion **510**) than width (e.g., dimension in the z-direction). Due to the flexibility of first flex portion **506**, first flex portion **506** may change shape according to external forces. For example, first flex portion **506** may bend, if desired, by user manipulation for example. In another example, first flex portion **506** may be altered (e.g., bent, twisted, etc.) according to a shape of an over mold applied over first flex portion **506**. Additionally, first flex portion **506** may be a single sided flexible PCB with conductive traces on one surface. Alternatively, the first flex portion **506** may be a double sided flexible PCB with conductive traces on both sides of first flex portion **506**. As such, a first plurality of conductive traces **505** may extend linearly along first flex portion **506** from first rigid/stiffener portion **503** to second rigid/stiffener portion **510** (e.g., parallel to the y-axis) on one or more surfaces of first flex portion **506**. The first plurality of conductive traces **505** may be conductive lines printed onto and/or etched into first flex portion **506**. Each one of the first plurality of conductive traces **505** may be electrically coupled to one of the plurality of wires **504**, for example via solder joints.

[0039] One or more emitters **514** (e.g., emitter unit **1200** of FIG. **12**) may be positioned on a surface (e.g., in the x-y plane and facing a positive z-direction) of second rigid/stiffener portion **510**. Similar to the first rigid/stiffener portion **503**, second rigid/stiffener portion **510** may be a stiffener of a flex type PCB with stiffeners, or a rigid portion of a rigid-flex-rigid PCB. Emitters **514** may be mounted on second rigid/stiffener portion **510** rather than a flex portion, such as first flex portion **506**, to prevent degradation of emitters **514**. Second rigid/stiffener portion **510** may maintain its shape under tension, such that bending of emitters **514** is prevented, thus preventing degradation or shape change of emitters **514** and/or couplings between emitters **514** and the rigid/stiffener portion and/or electric couplings between emitters **514** and conductive traces due to shape change of the PCB. Second rigid/stiffener portion **510** may be shaped as a circular disk, as shown in FIG. **5-6**, however, second rigid/stiffener portion **510** may also be other shapes including elliptical, rectangular, square, triangular, hexagonal, and the like.

[0040] In at least some examples, emitters **514** may emit light in the positive z-direction. More specifically, emitters **514** may be light-emitting diodes (LEDs). Emitters **514** may be electrically coupled to the first plurality of conductive traces **505** such that emitters **514** are electrically and/or communicatively coupled to the controller via the first plurality of conductive traces **505** and the plurality of wires **504**. Thus, emitters **514** may send and/or receive power and/or signals from the controller and/or the power source via the first plurality of conductive traces **505**. In this way, emitters **514** may be powered by the power source and emit light in response to signals from the controller, for example.

[0041] Second rigid/stiffener portion **510** may be connected to a first end of a second flex portion **508**, and a second end of second flex portion **508** may be connected to a third rigid/stiffener portion **512**. Similar to first flex portion **506**, second flex portion **508** may be a flex portion of a flex PCB with stiffeners, or a flex portion of a rigid-flex-rigid PCB. Thus, a shape of second flex portion **508** may change according to conditions described above. Second flex portion **508** may be ribbon shaped, similar to first flex portion **506**, with a thin thickness (e.g., in the z-direction) and a longer length (e.g., dimension in the x-direction) than width (e.g., dimension in the y-direction). When in the compact state as shown in FIG. **5**, second flex portion **508** may be wave shaped, and therein be or include a wave-shaped flexible PCB. For example, a cross section of second flex portion **508** in the x-z plane may be sinusoidal, and second flex portion **508** may extend laterally in the y-direction. The flex portion **508** may include a plurality of waves when in the compact state, with a distance **522** between each of the waves of the ribbon shape of the second flex portion **508**, wherein the distance **522** is not fixed such that the distance **522** may be increased when not in the compact state. The distance **522** may be parallel with the x-axis. In other examples, second flex portion **508** may be wave shaped but not sinusoidal, or have sharp peaks forming an accordion fold shape rather than a curved wave shape in the compact state. In yet other examples, second flex portion **508** may be twisted or spiraled, such as in a helical shape or a spring shape, rather than waved. In this way, the second flex portion may be shaped as a ribbon that may be adapted to change a distance **520** between second rigid/stiffener portion **510** and third rigid/stiffener portion **512**. The distance **520** may be parallel with the x-axis.

[0042] The second flex portion may be a single sided flexible PCB with conductive traces on one surface, or second flex portion **508** may be a double sided flexible PCB with conductive traces on both sides of second flex portion **508**. Accordingly, a second plurality of conductive traces **518** of the plurality of wires **504** may extend from second rigid/stiffener portion **510** to third rigid/stiffener portion **512**. Similar to the first plurality of conductive traces **505**, the second plurality of conductive traces **518** may be conductive lines printed onto and/or etched into second flex portion **508** on one or more surfaces of second flex portion **508**. In at least some examples, a number of the second plurality of conductive traces **518** may be different than (e.g., greater or less than) a number of the first plurality of conductive traces **505** as shown in FIG. **5**. However, the number of the first plurality of conductive traces **505** and the number of the second plurality of conductive traces **518** may be the same in other examples. In such examples, first flex portion **506** may be substantially the same as second flex portion **508**. The second plurality of conductive traces **518** may run parallel to the x-axis along second flex portion **508** such that the second plurality of conductive traces **518** are wave shaped in the x-z plane in the compact state.

[0043] One or more detectors **516** (e.g., detector unit **1203** of FIG. **12**) may be positioned on a surface (e.g., in the x-y plane and facing a positive z-direction) of third rigid/stiffener portion **512**. Third rigid/stiffener portion **512** may be a rigid portion of a rigid-flex-rigid PCB or a stiffener of a flex PCB with stiffeners. In either case, the rigidity (e.g., resistance to deformation or shape change under tension or other forces) of third rigid/stiffener portion **512** may prevent degradation to detectors **516** and/or the physical coupling of detectors **516** to third rigid/stiffener portion **512** and/or the electric couplings between the detectors and the second plurality of conductive traces **518** due to tension on the PCB. Similar to the second rigid/stiffener portion **510**, third rigid/stiffener portion **512** may be shaped as a circular disk, among other shapes as noted above according to a desired configuration. In at least some examples, the second rigid/stiffener portion **510** may be shaped approximately the same as third rigid/stiffener portion **512**.

[0044] In at least some examples, detectors **516** may be photodetectors capable of detecting light, for example from LEDs, when the light is aligned with detectors **516**. Additionally, detectors **516** may be able to send a corresponding electrical signal, for example to a controller which may interpret the electrical signal. Detectors **516** may be electrically coupled to the second plurality of conductive traces **518** such that detectors **516** may be powered, send signals, and/or receive signals via the second plurality of conductive traces **518**.

[0045] Because of the electric couplings between the plurality of wires **504** and first plurality of conductive traces **505**, between the plurality of conductive traces **505** and emitters **514**, between emitters **514** and the second plurality of conductive traces **518**, and between the second plurality of conductive traces **518** and detectors **516**, electric signals and/or power (e.g., current) may be transmitted between emitters **514**, detectors **516**, and the controller via the plurality of wires **504**, the first plurality of conductive traces **505**, and the second plurality of conductive traces **518**. In this way, detectors **516** and emitters **514** share electric couplings to the power source and controller. Thus, detectors **516** may be electrically coupled via emitters **514**, such that a separate electric coupling between detectors **516** and the power source and/or the controller is not demanded. In this way, in at least some examples, detectors **516** and emitters **514** may not be electrically coupled to each other via any other electric couplers than conductive traces (e.g., first plurality of conductive traces **505** and second plurality of conductive traces **518**) positioned therebetween. Thus, complexity of pulse oximetry hardware **500** may be reduced, and snagging and tangling of electric couplers (e.g., wires, power cords, etc.) may be prevented during operation of a stretchable sensor assembly in which pulse oximetry hardware **500** is incorporated.

[0046] In examples in which first rigid/stiffener portion **503**, first flex portion **506**, second rigid/stiffener portion **510**, second flex portion **508**, and third rigid/stiffener portion **512** are portions of a flex PCB with stiffeners, the stiffeners (e.g., first rigid/stiffener portion **503**, second rigid/stiffener portion **510**, and third rigid/stiffener portion **512**) may be applied to one or both sides of the flex PCB portions (e.g., first flex portion **506** and second flex portion **508**) for mechanical support, and the stiffeners may have an increased thickness than the flex PCB portions. In examples in which first rigid/stiffener portion **503**, first flex portion **506**, second rigid/stiffener portion **510**, second flex portion **508**, and third rigid/stiffener portion **512** are portions of a rigid-flex-rigid PCB, the rigid portions (e.g., first rigid/stiffener portion **503**, second rigid/stiffener portion **510**, and third rigid/stiffener portion **512**) may serve as mechanical support and an interface for electrical couplings, and may be formed integrally with the flex portions (e.g., first flex portion **506** and second flex portion **508**).

[0047] Pulse oximetry hardware **500** may be a non-limiting example of hardware incorporated in a stretchable sensor assembly, such as stretchable sensor assembly **700** of FIGS. **7-11** described below. In other examples, emitters **514** may be mounted on third rigid/stiffener portion **512** and detectors **516** may be mounted on second rigid/stiffener portion **510** such that emitters **514** are electrically coupled to cable **502** via detectors **516**.

[0048] FIG. **6** shows pulse oximetry hardware **500** with second flex portion **508** in a stretched state such that a distance **620** between second rigid/stiffener portion **510** and third rigid/stiffener portion **512** is greater than the distance **520** in the compact state, shown in FIG. **5**, between second rigid/stiffener portion **510** and third rigid/stiffener portion **512**. Second flex portion **508** may be adjusted such that the distance **522** between waves in the compact state as shown in FIG. **5** is increased to a larger distance **622**. The distance between the waves may further increase from the larger distance **622** until an unwaved shape (e.g., flat in an x-y plane) is reached for the second flex portion **508**. When in an unwaved shape, the second flex portion **508** may be at a maximum distance between the second rigid/stiffener portion **510** and third rigid/stiffener portion **512**, where the maximum distance is greater than the distance **620**. In this way, the distance between emitters **514** and detectors **516** may be increased in the stretched state compared to the compact state shown in FIG. **5**. Therefore, the distance between emitters **514** and detectors **516** may be variable. Thus, a stretchable sensor assembly (e.g., stretchable sensor assembly **700** of FIGS. **7-11**) including pulse oximetry hardware **500** may be adjusted to fit a range of target sizes adequately to allow alignment of emitters **514** and detectors **516** as is shown in FIGS. **10-11** and discussed further below.

[0049] Turning to FIG. **7**, it shows a stretchable sensor assembly **700**. Stretchable sensor assembly **700** includes pulse oximetry hardware **500** introduced with reference to FIG. **5**, and parts are labeled accordingly in FIG. **7**. Pulse oximetry hardware **500** may be over molded with an elastic cover **702** such that pulse oximetry hardware **500** is encased in elastic cover **702**. Elastic cover **702** may be an elastomer (e.g., natural rubber, polyurethane, polybutadiene, silicone, a mixture thereof, and the like) or other elastic material formed in the compact state, such as shown in FIG. **5**. Therefore, the compact state may be a “resting” state which elastic cover **702** may return to in the absence of tension (e.g., stretching or compressing force) due to the nature of the elastic material.

In this way, the ability to stretch from the compact state according to resistance of elastic cover **702** to deformation may allow for the stretchable sensor assembly to fit around a range of target sizes.

[0050] Briefly referring to FIGS. **5** and **7**, elastic cover **702**, which may envelop pulse oximetry hardware **500**, may include a first portion **701**, a second portion **714**, a third portion **724**, a fourth portion **716**, a fifth portion **722**, and a sixth portion **708**. First portion **701** may be adapted to cover first rigid/stiffener portion **503** and first flex portion **506**, second portion **714** may be adapted to at least partially cover second rigid/stiffener portion **510**, third portion **724** may be adapted to at least partially cover second flex portion **508**, and fourth portion **716** may be adapted to at least partially cover third rigid/stiffener portion **512**. Fifth portion **722** may be shaped substantially the same as third portion **724** such that fifth portion **722** and third portion **724** are each a wave-shaped band, although fifth portion **722** may not cover any parts of pulse oximetry hardware **500**, in at least some examples. Additionally, fifth portion **722** and third portion **724** may extend lengthwise perpendicular to one another. Sixth portion **708** may also not cover any parts of pulse oximetry hardware **500**, in at least some examples.

[0051] Second portion **714** may include a first opening **734** in which a first clear cover **704** may be placed. Likewise, fourth portion **716** may include a second opening **732** in which a second clear cover **706** may be placed. The first clear cover **704** and the second clear cover **706** may be transparent so as not to interfere with transmitting light from emitters **514** to detectors **516** when the stretchable sensor assembly is positioned on a target as shown in FIGS. **10-11**.

[0052] Sixth portion **708** may be used to secure stretchable sensor assembly **700** in a ring shape when positioned around a target, such as shown in FIGS. **10-11**. For example, sixth portion **708** may have a complementary counterpart (not shown) located on second portion **714**, such as to physically couple the second portion **714**. Sixth portion **708** and the complementary counterpart may fit together in a variety of ways without departing from the scope of this disclosure. For example, the complementary counterpart may be located on a bottom **710** of second portion **714** and may extend through a hole **712** in sixth portion **708**, interlocking with the hole **712** such that stretchable sensor assembly **700** forms a closed loop, or ring, around the target. In this way, elastic cover **702** may couple to itself to form a ring shape. In other examples, second portion **714** may be connected to sixth portion **708** by other means, such as snap-fit, hook and loop fasteners, and so on. By connecting second portion **714** and sixth portion **708**, emitters **514** and detectors **516** may be positioned oppositely across the target. Further, the waves in third portion **724** and fifth portion **722** may align emitters **514** and detectors **516** by ensuring equidistance of spacing between emitters **514** and detectors **516** around the ring shape regardless of the target size (so long as the target size is within a target size range), as is discussed further below.

[0053] Stretchable sensor assembly **700** may be a non-limiting example of a stretchable sensor assembly, and other arrangements of elastic cover portions may be used without departing from the scope of this disclosure. For example, rather than forming an L-shape as shown in FIG. **7**, a T-shape may be formed. To elaborate, a waved portion may be positioned on either side of second portion **714**, rather than on either side of fourth portion **716**. For example, a stretchable sensor assembly may include first portion **701**, second portion **714**, third portion **724**, and fourth portion **716** as shown in FIG. **7**, but fifth portion **722** may be positioned to the right (e.g., in a negative x-direction) of second portion **714**, and sixth portion **708** may be positioned further to the right than fifth portion **722**. In this example, sixth portion **708** may be adapted to couple to fourth portion **716** to form a ring shape that may be fitted to a target in a similar manner to the stretchable sensor assembly **700**.

[0054] Referring to FIGS. **8** and **9**, side cross section views (e.g., cross sections in an x-y plane) of stretchable sensor assembly **700** in a compact state **800** and a stretched state **900**, respectively, are shown. As noted above, third portion **724** and fifth portion **722** may be approximately the same shape (e.g., wave shape) and material, thus having approximately the same resistance to deformation. Consequently, in the compact state **800**, a length **720** of third portion **724** may be approximately the same as a length **718** of fifth portion **722**. For example, the length **720** and the length **718** may result in a first distance between second portion **714** and fourth portion **716**, and between fourth portion **716** and sixth portion **708**. Likewise, in the stretched state, a length **920** of third portion **724** may be approximately the same as a length **918** of fifth portion **722**. For example, the length **920** and the length **918** may result in a second distance between second portion **714** and fourth portion **716**, and between fourth portion **716** and sixth portion **708**. The length **920** and the length **918** may be larger than the length **720** and the length **718**. In this way, when stretchable sensor assembly **700** is positioned around a subject, such as a finger, emitters **514** and detectors **516** may be aligned as is further described below.

[0055] Further, the ability to stretch third portion **724** and fifth portion **722** allows for alignment of emitters **514** and detectors **516** around a range of target sizes (e.g., finger sizes). For example, the compact state **800** may be sized appropriately for a first small target that is smaller relative to a standard target, such as a first finger of a pediatric patient, while the stretched state **900** may be able to fit a second large target that is larger relative to the standard target or the smaller target, such as a second finger of an adult patient. Another state may exist between the compact and stretched states which may fit a target of size between the sizes of the first small target and second large target, such as a finger larger than the first finger and the smaller than the second finger. Likewise, another state may exist above the compact and stretched states, such as a maximum stretched state, which may fit a target larger than the second large target, such as a finger or another feature of a subject, such as a thumb, larger than the second finger. Due to the third portion **724** and fifth portion **722** having approximately the same wave shape, emitters **514** and detectors **516** may be arranged equidistantly around the ring shape formed when positioned around, or fit to, a target. In this way, a range of target sizes (e.g., finger sizes) may be accommodated by stretchable sensor assembly **700**, with emitters **514** and detectors **516** aligned for accurate measurement, for example of blood oxygen saturation.

[0056] The wave shape described herein is a non-limiting example of a stretchable shape that an elastic cover may take between emitters and detectors of a stretchable sensor assembly, such as third portion **724** and fifth portion **722** of stretchable sensor assembly **700**. In other examples, the stretchable shape may have sharp corners bent similar to an accordion fold rather than smooth bends of a wave shape. In other examples, the stretchable shape may be a twist or spiral, similar to a spring shape. In any examples, the stretchable shape may allow the distance between the emitters and detectors to be varied according to the size of a target, such that the stretchable sensor assembly may fit a range of target sizes with the emitters and detectors aligned for accurate measurements.

[0057] In the example where the stretchable shape is a wave shape, the resting shape taken in a relaxed state of the elastic cover may be a wave, and the maximally stretched state may be a flat ribbon. For example, second flex portion **508** of pulse oximetry hardware **500** may be deformed from a flat ribbon shape to a wave shape prior to being over molded with elastic cover **702**. As a result, when the elastic material is molded onto the wave shaped second flex portion **508**, elastic cover **702** takes a wave shape in third portion **724**. A mold used to over mold pulse oximetry hardware **500** may also include an approximately same wave shape to form fifth portion **722** as is used to form third portion **724**. Thus, elastic cover **702** takes the wave shape in third portion **724** and fifth portion **722** as a resting shape which elastic cover **702** takes in an absence of tension or compression. Thus, elastic cover **702** holds second flex portion **508** (shown in FIG. **5**) in a wave shape in an absence of tension. Manipulation of elastic cover **702** by a user to fit stretchable sensor assembly **700** to a target, such as a finger, may change the shape of elastic cover **702**, thereby changing a shape of pulse oximetry hardware **500**, more specifically second flex portion **508**. On its own, pulse oximetry hardware **500** may not be elastic such that pulse oximetry hardware returns to the compact state after being stretched. Thus, the ability to stretch stretchable sensor assembly **700** may be provided by material characteristics and the resting shape of elastic cover **702**, while flex portions of pulse oximetry hardware **500** may be adjusted according to behavior of elastic cover **702**.

[0058] In this way, stretchable sensor assembly **700** may fit a range of target sizes, for example finger sizes, such that stretchable sensor assembly is in face-sharing contact with the target in at least two portions (e.g., second portion **714** and fourth portion **716**) and the emitters and detectors are aligned. A first ring formed by stretchable sensor assembly **700** in the relaxed state (e.g., without tension), or compact state, may serve as a lower threshold for the target size. For example, the target diameter may not be significantly smaller than a first inner diameter of the first ring. A second ring formed by stretchable sensor assembly **700** in a stretched state in which the waved portions are flattened may serve as an upper threshold for the target size. For example, the target diameter may not be greater than a second inner diameter of the second ring. Thus, the target diameter may be between and including the lower threshold and upper threshold, for example. In other examples, a range of target sizes may be determined by other measurements than diameters, such as circumferences or perimeters.

[0059] Turning to FIGS. **10** and **11**, stretchable sensor assembly **700** is shown positioned on (e.g., worn on) a target **1002** in a top view **1000** and a front view **1100**, respectively. When positioned on the target **1002**, the stretchable sensor assembly **700** may extend about, such as around, and be in surface sharing contact with the target **1002**. Therefore, stretchable sensor assembly may be a wearable system. In an example in which emitters **514** and detectors **516** (as shown in FIG. **5**) are light emitters and photodetectors, respectively, stretchable sensor assembly **700** may be considered a blood oxygen sensor. Thus, stretchable sensor assembly **700** may also be a wearable blood oxygen sensor, in at least some examples. In such examples, the target **1002** may be a finger of a patient. Reference axes **1004** are provided in FIGS. **10** and **11**, including an x-axis, a y-axis, and a z-axis. For example, the z-axis may be parallel to the path of light emitted from the emitters of stretchable sensor assembly **700**. Additionally or alternatively, the x-axis may be parallel to an axial length of the target **1002**, and the y-axis and z-axis may be parallel to radial directions of the target **1002**.

[0060] As described above and shown in FIGS. **10-11**, sixth portion **708** may be connected to second portion **714** when positioned on a target, such as the target **1002**. Sixth portion **708** is shown connected on top (e.g., in a positive z-direction) of second portion **714** via the connecting member **1006** which may be a complementary counterpart to the hole **712** in sixth portion **708**. However, in other examples, the sixth portion may connect to the second portion by other means and in other orientations. For example, a side of sixth portion **708** may be connected to a side of second portion **714** rather than the top. In at least some examples, the complementary counterpart may removably lock sixth portion **708** and second portion **714** together. In other examples, sixth portion **708** and second portion **714** may be permanently linked together, such that the ring shape may not be unfolded such as shown in FIGS. **8** and **9**. Thus, a variety of connection mechanisms may be employed in order to position a stretchable sensor assembly around a target without departing from the scope of this disclosure.

[0061] Due to the connection between sixth portion **708** and second portion **714**, emitters (e.g., emitters **514** of FIGS. **5-9**) and detectors (e.g., detectors **516** of FIGS. **5-9**) positioned within second portion **714** and fourth portion **716**, respectively, may be aligned such that an accurate oxygen saturation level may be determined. As described above in reference to FIGS. **8** and **9**, third portion **724** and fifth portion **722** may have approximately the same wave shape. Therefore, third portion **724** and fifth portion **722** may have approximately the same resistance to deformation, thus allowing for approximately equal distances between second portion **714** and fourth portion **716**, and between fourth portion **716** and sixth portion **708** in any state (e.g., compact state **800** in FIG. **8**, stretched state **900** in FIG. **9**, positioned on the target **1002**). Therefore, distances between the emitters and detectors around the target **1002** in either direction (e.g., clockwise and counterclockwise in FIG. **11**) may be approximately the same as each other, regardless of the size (e.g., diameter) of the target **1002**. Consequently, the emitters and detectors may be aligned such that light emitted from the emitters may follow a path shown by an arrow **1102** through the target and be intercepted by the detectors.

[0062] The technical effect of the stretchable sensor assembly disclosed herein is to ensure alignment of sensor components, such as emitters and detectors, when positioned around a target, for a range of sizes of the target. Additionally, the stretchable sensor assembly disclosed herein avoids having separate electric couplings for the detectors and emitters which prevents tangling and/or snagging of extra wires/cords and may reduce resource demand.

[0063] The disclosure also provides support for a device, comprising: an emitter, a detector, a power/signal cable directly connected to one of the detector or the emitter, and a wave-shaped flexible printed circuit board (PCB) disposed between the emitter and the detector, wherein the wave-shaped flexible PCB includes a first plurality of conductive traces. In a first example of the system, the emitter is a light emitting diode and the detector is a photodetector. In a second example of the system, optionally including the first example, a second flexible PCB including a second plurality of conductive traces is disposed between the power/signal cable and the emitter. In a third example of the system, optionally including one or both of the first and second examples, the wave-shaped flexible PCB is perpendicular to the second flexible PCB. In a fourth example of the system, optionally including one or more or each of the first through third examples, the device is over molded with an elastic cover that holds the wave-shaped flexible PCB in a wave shape in an absence of tension and allows the wave-shaped flexible PCB to be stretched under tension. In a fifth example of the system, optionally including one or more or each of the first through fourth examples the elastic cover further comprising a wave-shaped band extending from the detector and configured to lock onto a portion comprising the emitter. In a sixth example of the system, optionally including one or more or each of the first through fifth examples, the wave-shaped band has approximately the same wave shape as the wave-shaped flexible PCB. In a seventh example of the system, optionally including one or more or each of the first through sixth examples, the emitter and detector are electrically coupled via the conductive traces therebetween, such that the detector is electrically coupled to the power/signal cable via the emitter.

[0064] The disclosure also provides support for a wearable system, comprising: a sensor comprised of an emitter coupled to a detector via a wave-shaped flexible printed circuit board (PCB), the wave-shaped flexible PCB including conductive traces, an elastic material molded over the wave-shaped flexible PCB adapted to maintain a first distance between the emitter and detector when not under tension. In a first example of the system, the elastic material couples to itself to form a ring shape. In a second example of the system, optionally including the first example, tension increases the first distance to a second distance. In a third example of the system, optionally including one or both of the first and second examples, the emitter and the detector are electrically coupled to each other by the conductive traces, where there are no other electric couplers than the conductive traces therebetween. In a fourth example of the system, optionally including one or more or each of the first through third examples, the emitter and detector are positioned equidistantly along the ring shape. In a fifth example of the system, optionally including one or more or each of the first through fourth examples, the emitter is a light emitting diode and the detector is a photodetector. In a sixth example of the system, optionally including one or more or each of the first through fifth examples, the emitter is mounted on a first rigid PCB and the detector is mounted on a second rigid PCB, the wave-shaped flexible PCB is disposed between the first rigid PCB and the second rigid PCB.

[0065] The disclosure also provides support for a wearable blood oxygen sensor, comprising: a light emitter coupled to a first rigid printed circuit board (PCB), a photodetector coupled to a second rigid PCB, a flexible PCB, comprising conductive traces coupled at a first end to the light emitter and at a second end to the photodetector, the flexible PCB adapted to change shape to adjust a distance between the light emitter and the photodetector. In a first example of the system, the wearable blood oxygen sensor is over molded with an elastic cover that changes the shape of the flexible PCB under tension to adjust the distance between the light emitter and the photodetector. In a second example of the system, optionally including the first example, the flexible PCB is wave shaped when not under tension. In a third example of the system, optionally including one or both of the first and second examples, the wearable blood oxygen sensor is wearable on a finger. In a fourth example of the system, optionally including one or more or each of the first through third examples, the wearable blood oxygen sensor fits around a range of finger sizes such that the light emitter and the photodetector are aligned.

[0066] As used herein, an element or step recited in the singular and preceded with the word “a” or “an” should be understood as not excluding plural of said elements or steps, unless such exclusion is explicitly stated. Furthermore, references to “one embodiment” of the present invention are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Moreover, unless explicitly stated to the contrary, embodiments “comprising,” “including,” or “having” an element or a plurality of elements having a particular property may include additional such elements not having that property. The terms “including” and “in which” are used as the plain-language equivalents of the respective terms “comprising” and “wherein.” Moreover, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements or a particular positional order on their objects.

[0067] FIGS. **1-12** show example configurations with relative positioning of the various components. If shown directly contacting each other, or directly coupled, then such elements may be referred to as directly contacting or directly coupled, respectively, at least in one example. Similarly, elements shown contiguous or adjacent to one another may be contiguous or adjacent to each other, respectively, at least in one example. As an example, components laying in face-sharing contact with each other may be referred to as in face-sharing contact. As another example, elements positioned apart from each other with only a space there-between and no other components may be referred to as such, in at least one example. As yet another example, elements shown above/below one another, at opposite sides to one another, or to the left/right of one another may be referred to as such, relative to one another. Further, as shown in the figures, a topmost element or point of element may be referred to as a “top” of the component and a bottommost element or point of the element may be referred to as a “bottom” of the component, in at least one example. As used herein, top/bottom, upper/lower, above/below, may be relative to a vertical axis of the figures and used to describe positioning of elements of the figures relative to one another. As such, elements shown above other elements are positioned vertically above the other elements, in one example. As yet another example, shapes of the elements depicted within the figures may be referred to as having those shapes (e.g., such as being circular, straight, planar, curved, rounded, chamfered, angled, or the like). Further, elements shown intersecting one another may be referred to as intersecting elements or intersecting one another, in at least one example. Further still, an element shown within another element or shown outside of another element may be referred as such, in one example. FIGS. **5-11** are shown approximately to scale, although other relative dimensions may be used.

[0068] This written description uses examples to disclose the invention, including the best mode, and also to enable a person of ordinary skill in the relevant art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those of ordinary skill in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A device, comprising: an emitter; a detector; a cable directly connected to one of the detector or the emitter; and a wave-shaped flexible printed circuit board (PCB) disposed between the emitter and the detector, wherein the wave-shaped flexible PCB includes a first plurality of conductive traces.

2. The device of claim 1, wherein the emitter is a light emitting diode and the detector is a photodetector.

3. The device of claim 1, wherein a second flexible PCB including a second plurality of conductive traces is disposed between the cable and the emitter.

4. The device of claim 3, wherein the wave-shaped flexible PCB is perpendicular to the second flexible PCB.

5. The device of claim 1, wherein the device is over molded with an elastic cover that holds the wave-shaped flexible PCB in a wave shape in an absence of tension and allows the wave-shaped flexible PCB to be stretched under tension.

6. The device of claim 5, the elastic cover further comprising a wave-shaped band extending from the detector and configured to lock onto a portion comprising the emitter, or extending from the emitter and configured to lock onto a portion comprising the detector.

7. The device of claim 6, wherein the wave-shaped band has approximately the same wave shape as the wave-shaped flexible PCB.

8. The device of claim 1, wherein the emitter and detector are electrically coupled via the first plurality of conductive traces therebetween, such that the detector is electrically coupled to the cable via the emitter.

9. A wearable system, comprising: a sensor comprised of an emitter coupled to a detector via a wave-shaped flexible printed circuit board (PCB), the wave-shaped flexible PCB including conductive traces; and an elastic material molded over the wave-shaped flexible PCB adapted to maintain a first distance between the emitter and detector when not under tension.

10. The wearable system of claim 9, wherein the elastic material couples to itself to form a ring shape.

11. The wearable system of claim 9, wherein tension increases the first distance to a second distance.

12. The wearable system of claim 9, wherein the emitter and the detector are electrically coupled to each other by the conductive traces, where there are no other electric couplers than the conductive traces therebetween.

13. The wearable system of claim 10, wherein the emitter and detector are positioned equidistantly along the ring shape.

14. The wearable system of claim 9, wherein the emitter is a light emitting diode and the detector is a photodetector.

15. The wearable system of claim 9, wherein the emitter is mounted on a first rigid PCB and the detector is mounted on a second rigid PCB, the wave-shaped flexible PCB is disposed between the first rigid PCB and the second rigid PCB.

16. A wearable blood oxygen sensor, comprising: a light emitter coupled to a first rigid printed circuit board (PCB); a photodetector coupled to a second rigid PCB; and a flexible PCB, comprising conductive traces coupled at a first end to the light emitter and at a second end to the photodetector, the flexible PCB adapted to change shape to adjust a distance between the light emitter and the photodetector.

17. The wearable blood oxygen sensor of claim 16, wherein the wearable blood oxygen sensor is over molded with an elastic cover that changes the shape of the flexible PCB under tension to adjust the distance between the light emitter and the photodetector.

18. The wearable blood oxygen sensor of claim 16, wherein the flexible PCB is wave shaped when not under tension.
19. The wearable blood oxygen sensor of claim 16, wherein the wearable blood oxygen sensor is wearable on a finger.
20. The wearable blood oxygen sensor of claim 19, wherein the wearable blood oxygen sensor fits around a range of finger sizes such that the light emitter and the photodetector are aligned.